

From this week's labs to your Azure tenant

Every lab this week runs on Colab and a vendor API key, so it works on any laptop. This page says what replaces each piece in an Azure tenant. Names and facts checked on learn.microsoft.com on 27 September 2026; Azure renames services often, so check the link before you build.

The mapping

Day	In the lab	In Azure	Check first
1 to 5	Colab notebook	Azure Machine Learning compute instance: managed Jupyter or VS Code, GPU sizes available	Network must allow websockets to the Azure ML instance hosts
1 to 5	OpenAI API key	Azure OpenAI in Microsoft Foundry Models. The same OpenAI Python SDK works: base URL <a href="https://<resource>.openai.azure.com/openai/v1/">https://<resource>.openai.azure.com/openai/v1/	Which models and deployment types your region offers (Foundry portal, Quota)
1, 3	Build, buy or host option B	Deployment types: Global Standard (any region), Data Zone Standard (US, EU or APAC only), Standard (your region)	There is no Middle East data zone
2	Ollama on a Colab T4	A GPU VM you run (NVadsA10_v5, NCasT4_v3) serving vLLM, or an Azure ML managed online endpoint	GPU quota and which sizes your region has
2	Production serving engine	vLLM on that VM, or managed compute in Foundry (preview, A100 and H100 class GPUs)	Managed compute has no built-in content filtering
2	LoRA fine-tune on a T4	Fine-tuning in Microsoft Foundry: gpt-4.1-mini and others, plus some open models	Training runs in US or EU regions only; an idle fine-tuned deployment is deleted after 15 days
2, 4	Eval script and scoring rubric	Foundry evaluations: groundedness, relevance and custom evaluators	Keep your own held-out set: the evaluator is not the data
3	ChromaDB and BM25 hybrid search	Azure AI Search: hybrid (keyword plus vector) and the semantic ranker	Semantic ranker has a free monthly quota, then is billed
3	PDF and scanned page ingestion	Azure Document Intelligence in Foundry Tools; Azure Content Understanding	Test on your own scanned pages
4	LangGraph agent	Foundry Agent Service (hosts LangGraph and other frameworks)	Available in UAE North per Microsoft's region table
4	Guardrails and injection test	Azure AI Content Safety and Prompt Shields (direct and indirect prompt attacks)	Configured in Foundry as guardrails
4, 5	Traces and logs	Foundry tracing to Azure Monitor Application Insights (OpenTelemetry)	Ingestion is billed
4, 5	Custom MCP server over the ERP API	Azure API Management MCP servers (expose a REST API as MCP tools, govern existing servers); Azure Functions MCP extension	API Management supports MCP tools only, and not on the Consumption tier
5	Governance pack	Microsoft Entra ID managed identities, Microsoft Purview DSPM, Defender for AI Services	Map each governance template to the control that already exists

Three facts for OQ planning

- **Residency.** Data Zone deployments exist only for the US, EU and APAC. For in-region processing you need regional Standard or Provisioned deployments, and model availability differs by region. Check the Foundry portal before you promise residency.

- **Fine-tuning leaves the region.** Azure fine-tuning training runs in US and EU regions. If the training data may not leave the country, the self-hosted adapter from Day 2 is the path that stays in your tenant and region.
- **Models retire.** Azure publishes a retirement schedule per model; some gpt-4.1 versions are due to retire around mid October 2026. Plan the replacement on the day you choose a model (Day 1, S2).

Links

- Azure OpenAI v1 API: learn.microsoft.com/azure/foundry/openai/api-version-lifecycle
- Deployment types and data zones: learn.microsoft.com/azure/foundry/foundry-models/concepts/deployment-types
- Fine-tuning in Foundry: learn.microsoft.com/azure/foundry/openai/how-to/fine-tuning
- Model retirement schedule: learn.microsoft.com/azure/foundry/openai/concepts/model-retirement-schedule
- Azure AI Search hybrid search: learn.microsoft.com/azure/search/hybrid-search-overview
- API Management MCP servers: learn.microsoft.com/azure/api-management/mcp-server-overview